FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Main)-2020

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. D

Sol. From the property of vector product, we notice that **C** must be perpendicular to the plane formed by vector **A** and **B**. Thus **C** must be perpendicular to **A** and to **B**. Thus the statements (1) and (2) are correct. Now **A** + **B** vector also must lie in the plane formed by vector **A** and **B**. Thus **C** must be perpendicular to **A** + **B** also the cross product (**A** × **B**) gives a vector **C** which cannot be perpendicular to itself. Thus the statement (4) is wrong.

2. C

Sol. The force acting on the particle = $\frac{mdv}{dt}$

Power of the force = $\left(\frac{mdv}{dt}\right) v = k$ (constant)

$m \frac{v^2}{2} = kt + c$ ...(1)

at $t = 0, v = u$ $\therefore c = \frac{mu^2}{2}$

Now from (1),

$m \frac{v^2}{2} = kt + \frac{mu^2}{2}$

$\frac{1}{2} m (v^2 - u^2) = kt$ ...(2)

Again $\frac{mdv}{dt} v = k$

$m.v \frac{dv}{dx} v = k$

$\int_u^v mv^2 dv = \int_0^x k\,dx$

Intergrating, $\frac{1}{3}$ m ($v^3 - u^3$) = kx ...(3)

From eqn (2) and (3),

t = $\frac{3}{2}\left(\frac{v^2-u^2}{v^3-u^3}\right)$ x

3. A

Sol. From ohm's law
Electric field α current density

4. D

Sol. $e = \int_{-\ell/2}^{\ell/2} B\omega r dr = 0$

5. B

Sol. When current is maximum $= \frac{di}{dt} = 0$

6. D

7. A

Sol. Saturation for A and B are different hence intensities are different. Stopping potentials for A and B are equal hence frequencies are equal.

8. B

Sol. Characteristic wavelengths shift of left if z is increased hence B is incorrect

9. D

10. B

Sol. $\rho_1 g h_1^2 = \rho_2 g h_2$

$\frac{h_1}{h_2} = \frac{\rho_2}{\rho_1}$

11. B

12. A

Sol. Process A $\rightarrow$ B is isothermal
Process B $\rightarrow$ C isochoric
Process C $\rightarrow$ A is isobaric

13. C

Sol. The process described is isochoric
Therefore molar heat capacity $= C_v = 2.5R$

14. A

15. B

Sol. Capacitors are in series therefore

$\frac{C_1}{C_2} = \frac{V_2}{V_1} = \frac{2}{3}$

16. B

17. A

Sol. $mu = 3mv$

$V = u/3$

$e = -\dfrac{2u/3 - u/3}{-u} \Rightarrow e = 1/3$

18. D

Sol. Friction will be absent

$T = ma$

$mg - T = 2ma$

Acceleration of $a = g/3$

Acceleration of block $= 2g/3$

19. B

Sol. Tension at bottom end of rope = $T_1 = 2\times9.8$ N

$\because$ weight of rope acts on gravity centre,

Therefore, tension at upper end of rope,

$T_2 = (6 + 2) \times 9.8 = 8 \times 9.8$ N

Thus, $T_2 = 4T_1$

if v_1 & v_2 are respective velocity at bottom & upper end, then $v_1 = \sqrt{\frac{T_1}{m}}$ & $v_2 = \sqrt{\frac{T_2}{m}}$

$\therefore v_2 = 2v_1$ ($\because T_2 = 4T_1$)

Frequency n does not depend on medium,

therefore $v \propto \lambda$

if λ_1 & λ_2 are respective wavelength at bottom & upper end

$\lambda_2 = 2\lambda_1 = 2 \times 0.06 = 0.12$ m

20. C

Sol. The fringe–width β is given by

$\beta = \dfrac{\lambda D}{2d}$ and $\beta' = \dfrac{\lambda D'}{2d}$

where λ is the wavelength of light used, D is the distance of the screen from the two slits and 2 d is the separation between two slits.

Now $\beta - \beta' = \dfrac{\lambda(D - D')}{2d}$

$\Rightarrow \lambda = \dfrac{(\beta - \beta')2d}{D - D'}$

$\lambda = \dfrac{(3\times10^{-5})(10^{-3})}{5\times10^{-2}}$

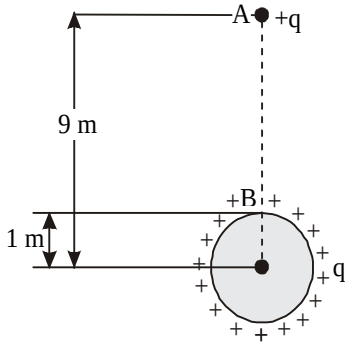
$= 0.6 \times 10^{-6}$ m = 6000 Å

SECTION – B

21. 6

22. 4

Sol. Keeping in mind that here both electric and gravitational potential energy are changing and for external point a charged sphere behaves as whole of its charge were concentrated at its centre, applying conservation of energy between initial and final position, we have



$$\frac{1}{4\pi\varepsilon_0}\frac{qq}{9} + mg \times 9 = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{1} + mg \times 1$$

or $q^2 = \dfrac{80\times10^{-3}\times9.8}{10^9}$

or $q = 28\ mC$

SECTION – C

23. 00019.00

24. 00080.00

Sol. Distance $AD = \dfrac{4}{5}\times100 = 80\,cm$

25. 00060.02

Sol. Since the scale is graduated at 10ºC,
1 cm of the scale at 10ºC = exactly 1 cm
$\therefore$ 1 cm of the scale at 30ºC
= exactly $(1 + 18 \times 10^{-6} \times 20)$cm
$\therefore$ 60 cm of the scale at 30ºC
= exactly $60.00\ (1 + 36 \times 10^{-5})$
= 60.02 cm

Chemistry

PART – II

SECTION – A

26. B

Sol. $\pi = i.CRT$

$$i = \frac{8.21}{0.15 \times 0.0821 \times 300} = 2.22$$

27. C

28. A

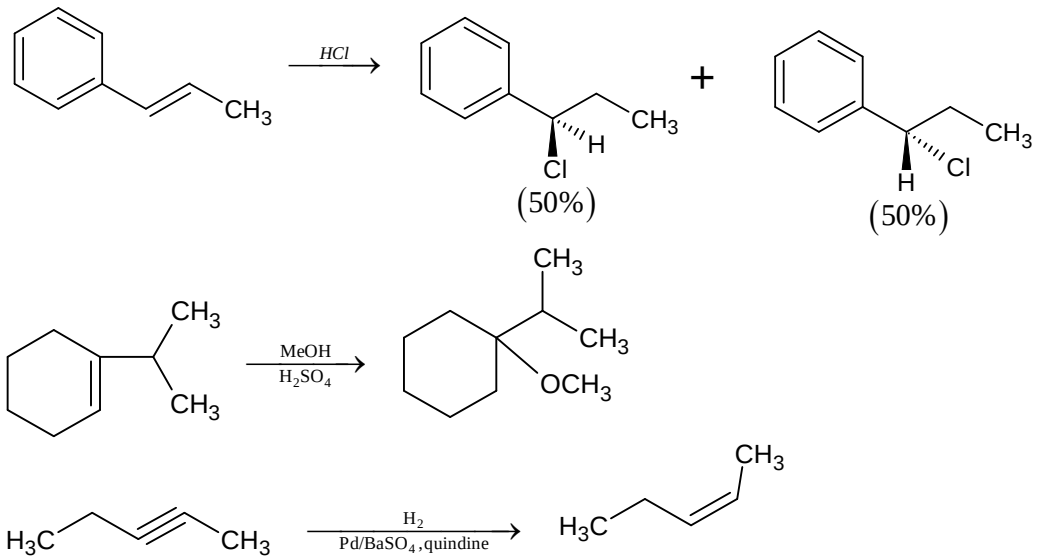
Sol. Let solubility barium sulphite is x mol/L and that of barium sulphate is y mol/L.

$(x+y)x = 8\times10^{-7}$

$(x+y)y = 1\times10^{-8}$

29. D

Sol.



30. C

31. A

32. B

Sol. Numbers of mole of A=10/150=1/15
Moles of B=0.20*(200/1000)=0.04
A is limiting reagent hence moles of X formed=(1/15)/2
P=(1/30)*0.0821*300/0.2=4.105atm

33. C

Sol. Due to inert pair effect, +2 oxidation state stability increases down the group in group 14. Hence element should be Pb(Lead).

34. D

Sol. CoS is black colour ppt.

35. B

36. A
Sol. A =XeF_6

37. B
Sol. Cerussite=$PbCO_3$

38. D
Sol. $Fe(SCN)_3$ is red colour

39. D
Sol. Brown ring complex is $[Fe(H_2O)_5(NO)]^{+2}$

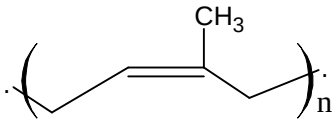
40. B

41. B
Sol. Conjugate base formed after removal of H (b) is resonance stabilized by both keto and alkene group.

42. C

43. A

44. D
Sol. AgCl is soluble in both liquid ammonia and in hypo solution

45. C
Sol.  is natural rubber

SECTION – B

46. 6
Sol. If ion has unpaired electron it is paramagnetic in nature

47. 4

SECTION – C

48. 00058.00
Sol. Y is allylic alcohol

49. 00284.90

Sol. Degree of dissociation $(\alpha) = \frac{\wedge_m}{\wedge_m^o}$

$\Rightarrow \wedge_m = \wedge_m^o \times \alpha$

$= \wedge_m^o \times \sqrt{\frac{K_a}{C}}$ (by Ostwald dilution law)

$= 390 \times \sqrt{\frac{1.8 \times 10^{-5}}{0.18}} = 3.90 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$

$$\text{Conductivity} = \frac{\text{molarity} \times \wedge_m}{1000}$$

$$= 7.02 \times 10^{-4} \text{ ohm}^{-1} \text{ cm}^{-1}$$

$$\text{Also conductivity} = \text{Cell constant} \times \frac{1}{\text{Resistance}}$$

$$\Rightarrow \text{ Resistance} = \frac{\text{Cell constant}}{\text{Conductivity}}$$

$$= \frac{0.2}{7.02 \times 10^{-4}} = 284.9 \text{ ohm}$$

50. 00009.74

Sol. $p^I = (8.95 + 10.53)/2$

Mathematics

PART – III

SECTION – A

51. C

Sol. Differential equation is $\frac{x^2}{a^2+\lambda}-\frac{y^2}{b^2+\lambda}=\frac{1}{4}$ $(\lambda \in R)$

Order is 1 as differential solution has 1 effective parameter λ.
Degree is 2 as highest power of effective parameter is 2 in differential solution.

52. B

Sol. $f(x) = 1 + x^2 + x^4 + x^6 + \ldots \infty \quad |x| \le 1$

$\Rightarrow f^n(0) = n!$, where n is even.

53. A

Sol. $f^r(x) = (x^n - 2r)^{1/n} \Rightarrow \int \left(f^6(x)\right)^n dx = \int (x^n - 12)dx = \frac{x^{n+1}}{n+1} - 12x + c$.

54. D

Sol. $\frac{d}{dx}(\sin kx) = k\cot kx \sin kx$

$\Rightarrow \frac{d}{dx}\left(\prod_{k=1}^{2n} \sin(kx)\right) = \sum_{k=1}^{2n} (k\cot kx) f(x)$.

55. B

Sol. $f'(-1)^- = -1$; $f'(-1)^+ = -2$

$\Rightarrow$ non-differentiable at $x = -1$.

56. B

Sol. Divide numerator and denominator by $\sqrt{x}$

$\Rightarrow$ given limit $= \frac{4}{\sqrt{4}} = 2$.

57. D

Sol. $\left(\lim_{n\to\infty}\left(\frac{n!}{n^n}\right)^{1/n}\right)^{\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{3n}\frac{r/n}{\sqrt{16+(r/n)^2}}} = \left(\frac{1}{e}\right)^{\int_0^3 \frac{x}{\sqrt{16+x^2}}dx} = \frac{1}{e}$.

58. B

Sol. Let $g(x) = \frac{2007}{x}$.

Let x decreases from k + 1 to k then g(x) increases from $\frac{2007}{k+1}$ to $\frac{2007}{k}$ $\Rightarrow$ The increase in the value of g(x) is $\frac{2007}{k(k+1)}$.

The increase is less than 1 if $k(k + 1) > 2007$

$\Rightarrow k = 45, 46, \ldots, 2007$.

$\Rightarrow$ g(x) does not miss any value as x decrease from 2007, 2006, … till 45.

Also $\left[\frac{2007}{45}\right] = 44$, $\left[\frac{2007}{44}\right] = 45$, $\left[\frac{2007}{43}\right] = 46$, $\left[\frac{2007}{42}\right] = 47$, $\left[\frac{2007}{41}\right] = 48$ and $\left[\frac{2007}{40}\right] = 50$

$\Rightarrow \left[\frac{2007}{x}\right]$ can never be equal to 49

$\Rightarrow$ the maximum value of k is 40.

59. A

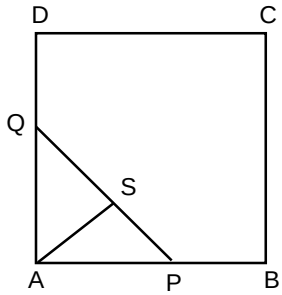
Sol. f(x) = 0 has minimum 2 solutions in [0, 4]
g(x) = 0 has minimum 3 solutions in [0, 4]
g′(x) = 0 has minimum 2 solutions in [0, 4]
f(x) g′(x) = 0 has minimum 4 solutions in [0, 4]
$\frac{d}{dx}(f(x)g'(x))$ = 0 has minimum 3 solutions in [0, 4].

60. D

Sol. $f(x)g(x) = 2$ …(a)
$\Rightarrow f'(x)g(x) + f(x)g'(x) = 0$ …(1)
$\Rightarrow f''(x)g(x) + 2f'(x)g'(x) + f(x)g''(x) = 0$
$\Rightarrow \frac{f''(x)g(x)}{f(x)g'(x)} + \frac{2f'(x)}{f(x)} + \frac{g''(x)}{g'(x)} = 0$ (dividing throughout by f(x)g′(x))
$\Rightarrow \frac{f''(x)}{f'(x)} - \frac{g''(x)}{g'(x)} = \frac{2f'(x)}{f(x)}$ $\left(\text{Replacing } \frac{g(x)}{f(x)g'(x)} = -\frac{1}{f'(x)} \text{ from}(1)\right)$
$\frac{2f'(x)}{f(x)}$ = 0 has no real roots since $f(x) \neq 0$ for any x. (from (a))

61. B

Sol. Let S be the mid point of PQ.
Since $\angle PAQ = \frac{\pi}{2}$, we get AS = SP = SQ = $\frac{1}{2}$
$\Rightarrow$ S lies on the quarter circle of radius $\frac{1}{2}$ with centre at A.
Similarly S can also lie on quarter circle of radius $\frac{1}{2}$ with centre at B, C or D
$\Rightarrow$ area A = $1 - \frac{\pi}{4}$.



62. C

Sol. Equation of tangent is $y - (t^2 - t + 2) = \frac{2t-1}{2t+1}(x - (t^2 + t + 2))$
Putting x = 2, y = 2, we get t = 0.

63. B

Sol. $I = \frac{1}{2}\int_a^b (f(x) + f(a+b-x))dx$
$= \frac{1}{2}\int_{-5}^{2} \frac{(6x^2+18x+18)}{x^2+3x+3}dx = 21$ $\left(f(x) = \frac{3x^2+x-3}{x^2+3x+3}\right)$

64. B

Sol. Let cost incurred to travel 200 kms be c(x)

$\Rightarrow c(x) = \left(100 + \frac{x^2}{60}\right)\frac{200}{x} \times 30 + 200 \times \frac{200}{x} = \frac{640000}{x} + 100x$

$\Rightarrow c'(x) < 0$ in $x \in [40, 70]$

$\Rightarrow c(x)$ is minimum for $x = 70$ in $x \in [40, 70]$.

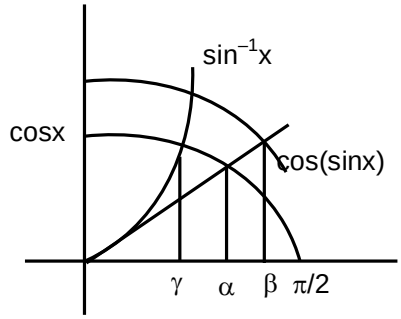
65. B

Sol. Equation (1) is $\cos x = x$ equation (2) is $\cos(\sin x) = x$ and equation (3) is $\cos x = \sin^{-1} x$

since $\sin x \le x \Rightarrow \cos(\sin x) \ge \cos x$

now drawing the graphs of the functions

we see that $\gamma < \alpha < \beta$.



66. A

Sol. $f'(x) = \lim_{h\to 0} \frac{f(x)\left(\frac{f(x+h)}{f(x)} - 1\right)}{h} = f(x) \lim_{h\to 0} \frac{f(h) - 1}{h}$ (Putting $x = x + h$, $y = x$)

$= f(x) f'(0)$ ($\because$ $f(0) = 1$ putting $x = y$)

$= pf(x)$

$\Rightarrow f(x) = e^{px}$

$f'(-5) = \frac{p}{e^{5p}} = \frac{p^2}{q}$ ($\because$ $f'(5) = pe^{5p} = q$).

67. C

Sol. $|\hat{r} \times \bar{a}|^2 = \sin^2 \alpha$

$\Rightarrow |\hat{r} \times \bar{a}|^2 + |\hat{r} \times \bar{b}|^2 + |\hat{r} \times \bar{c}|^2 = \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma$

$= 3 - (\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma) = 2$.

68. B

Sol $x_1 B\, x_2 B\, x_3 B \ldots B x_{2n+1}$

Let black balls are arranged 1st and between every black balls there are $(2n + 1)$ number of places in which white balls from 0 to $(n - 1)$ to be arranged.

Let x_i, $i = 1, 2, 3, \ldots, (2n + 1)$ denotes the number of balls at place denoted by $x_1, x_2, x_3, \ldots, x_{2n+1}$.

$\Rightarrow \sum_{i=1}^{2n+1} x_i = 2n$, where $0 \le x_i \le n - 1$, $i = 1, 2, 3, \ldots, 2n + 1$

$\Rightarrow$ Number of such arrangements = coefficients of x^{2n} in $(1 + x + x^2 + \ldots + x^{n-1})^{2n+1}$

$= {}^{2n+1}C_2 - {}^{2n+1}C_1\, {}^{3n}C_n\, (-1)^n + {}^{4n}C_{2n}$.

69. B

Sol. Let α, β be the roots

$\Rightarrow \alpha + \beta = -2|a|$ and $\alpha\beta = 4$.

Since $\alpha, \beta \in I$ hence possible values are $(\alpha, \beta) \equiv (-1, -4)$ or $(-2, -2)$

For a to be minimum $|a|$ should be maximum

$\Rightarrow |\alpha + \beta|$ should be maximum.

Hence roots are -1 and $-4 \Rightarrow a = -5/2$.

70. C

Sol. Any point on the line through P(a, 2) is $(a + r\cos\theta, 2 + r\sin\theta)$ and meet the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ at two distinct points A and D

$\Rightarrow \frac{(a+r\cos\theta)^2}{9}+\frac{(2+r\sin\theta)^2}{4}=1$

$\Rightarrow (4\cos^2\theta + 9\sin^2\theta)r^2 + 4(2a\cos\theta + 9\sin\theta)r + 4a^2 = 0.$

$\Rightarrow PA \cdot PD = \frac{4a^2}{4\cos^2\theta+9\sin^2\theta}$

also line meets the axes at B and C

$\Rightarrow PB \cdot PC = \frac{2a}{\sin\theta\cos\theta}$

since PA, PB, PC and PD are in G.P.

$\Rightarrow \frac{4a^2}{4\cos^2\theta+9\sin^2\theta} = \frac{2a}{\sin\theta\cos\theta}$

$\Rightarrow 2a\sin2\theta + 5\cos2\theta = 13$

$\Rightarrow -1 \le \frac{13}{\sqrt{4a^2+25}} \le 1$

$\Rightarrow a \le -6$ or $a \ge 6$.

SECTION – B

71. 0

Sol . $|A(\text{adj } A)|^{1/3} = |A| \Rightarrow \frac{d^2}{dx^2}(|A(\text{adj}A)|^{1/3}) = \begin{vmatrix} 3 & 2 & 1 \\ 12 & 12x & 12x^2 \\ 1 & a & a^2 \end{vmatrix} = 0$ at x = a.

= 0.

72. 5

Sol. The system has a nonzero solution

$\therefore |A| = 0$

$\begin{vmatrix} 2 & 3 & -1 \\ 3 & 2 & k \\ 4 & 1 & 1 \end{vmatrix} = 0$

The solution is $(2a, -3a, -5a)$

Least positive integer value of z is 5.

SECTION – C

73. 00000.00

Sol. $f(x) = \int_{\frac{\pi}{2}}^{x} (\cos t - 2\sin t)dt$

$f'(x) = \cos x - 2\sin x < 0$ in $\left[\frac{\pi}{2}, \pi\right]$

$\therefore f(x)$ is decreasing function

$\therefore$ Greatest value = $f\left(\frac{\pi}{2}\right) = 0$.

74. 00001.00

Sol. $dy = x\tan y(\sin y - 1)dx$

$\frac{dy}{dx} + x\tan y = x\tan y.\sin y$

$\cos y\frac{dy}{dx} + x\sin y = x\sin^2 y$

Let $\sin y = t$

$\frac{dt}{dx} + xt = xt^2$

$\frac{1}{t^2}\frac{dt}{dx} + x\left(\frac{1}{t}\right) = x$

$\frac{1}{t} = u$

$= -\frac{du}{dx} + xu = x$

$\frac{du}{dx} + (-x)u = -x$

$ue^{-\int xdx} = \int -xe^{-\int xdx}dx$

$ue^{-\frac{x^2}{2}} = -\int xe^{-\frac{x^2}{2}}dx \;= e^{-\frac{x^2}{2}} + c$

$u = 1 + ce^{\frac{x^2}{2}}$

$t = \frac{1}{1 + ce^{\frac{x^2}{2}}}$

$\sin y = \frac{1}{1 + ce^{\frac{x^2}{2}}}.$

75. 00001.60

Sol. Distance between the foci

$2ac = 4\sqrt{2}$

$ac = 2\sqrt{2}$

Product of distances of foci from tangent $= b^2$

$2 \times 6 = b^2$

$b^2 = a^2 - a^2e^2$

$12 = a^2 - 8$

$a^2 = 20$

$a = \sqrt{20}$

Major axis $2a = \sqrt{80}$

$[2a] = 8.$

$\frac{[x]}{5} = \frac{8}{5} = 1.6$